Jacob Johnson

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EDUCATION

Michigan State University **Expected Graduation May 2027**

Bachelor of Applied Engineering Sciences, Product Engineering Focus, Supply Chain Management Concentration
Society of Applied Engineering Sciences, Hollander Makerspace 3D Designer

Relevant Coursework: CAD and Engineering Graphics, Design and Product Development, Materials Engineering, Fabrication Manufacturing, Thermodynamics, Statics, Computational Programming, Product Marketing and Market Strategy, Sustainable Systems Engineering

TECHNICAL SKILLS

CAD and Engineering: AutoCAD, SolidWorks, Fusion360, Creo, Mastercam, Siemens NX
Programming and Analysis: Python, MATLAB, C, Java, Excel, ArcGIS
Manufacturing: CNC Machining, G-Code, 3-D Printing, Additive Manufacturing
Productivity Tools: Microsoft Office (Excel, PowerPoint, Word), Google Workspace, Zoom

ENGINEERING EXPERIENCE

MSU IPF *(Engineering and Facilities)* **December 2023 – June 2025**

Technical Assistant | CAD Design, Technical Drafting, AutoCAD, Field Engineering

- **Created and revised AutoCAD floor plans and technical drawings** using field measurements and existing documentation to maintain accurate building records.
- **Collaborated with engineers and project teams** to develop construction and renovation drawings while applying AutoCAD and Revit drafting standards.
- **Performed on-site field verification,** collecting dimensional and visual data with measurement equipment and translating findings into CAD drawings and ArcGIS documentation.

PROJECTS

Mechanical CAD Design & Assembly | *ME 280* **May 2026 – August 2026**

Siemens NX, Parametric Modeling, Engineering Drawings

- Created 3D parametric models and mechanical components in Siemens NX using constrained sketches, feature-based modeling, datum geometry, holes, threads, and other manufacturing features.
- Built multi-component mechanical assemblies using fit- and function-based constraints, verifying component alignment, interfaces, and intended mechanical motion.
- Produced detailed 2D engineering drawings with orthographic/section views, dimensions, tolerances, and manufacturing information from completed 3D models.

Product Design & Additive Manufacturing **September 2025 – May 2026**

Fusion 360, FDM 3D Printing, DFM

- Designed and prototyped functional consumer products in Fusion 360, translating real-world needs into manufacturable 3D CAD models.
- Optimized tolerances, geometry, material usage, and print orientation through iterative prototyping to improve fit, durability, and manufacturability.
- Managed the full concept → CAD → prototype → test → redesign cycle, incorporating test results into design revisions and final functional products.

Non-USGA Golf Club Face Prototype **April 2023 – May 2023**

Fabrication and Manufacturing, Mastercam, CNC Vertical Mill, G-Code

- Designed a custom aluminum golf club face with pyramidal groove geometry to reduce ball spin and produce a flatter trajectory.
- Developed 2.5D roughing and finishing toolpaths, generated G-code, and manufactured the prototype using a CNC vertical mill with dimensional verification against the CAD design.
- Conducted comparative range testing against a standard club face, documenting and analyzing ball-flight data to evaluate prototype performance.